

Cycle 3-Group 16

Gender and Current Alcohol Use

YRBS 2007 · N = 12,615

RESEARCH QUESTION

Primary

Is the proportion of current alcohol use different between male and female students?

Extended

Does the relationship between gender and current alcohol use vary across different age groups (12–18)?

GROUPS & VARIABLES

Independent: Sex (Male/Female), Age (12–18)
Dependent: Current Alcohol Use

METHOD

Two-Proportion Z-Test (Primary RQ) → upgraded to
Chi-Square Tests of Independence
upon introducing Age (12–18) as a multi-level variable.

GENDER

$$\chi^2(1) = 1.82$$
$$p = 0.178$$

Not Significant X

AGE

$$\chi^2(6) = 202.41$$
$$p < 0.0001$$

Highly Significant ✓

AGE TREND (14–18)

Alcohol use rises steadily with age — from ~32.7% at age 14 to ~53.0% at age 18 (averaged across both sexes).

GENDER × AGE CROSSOVER

Age 14–15: Female > Male (38.1% vs 27.2%)

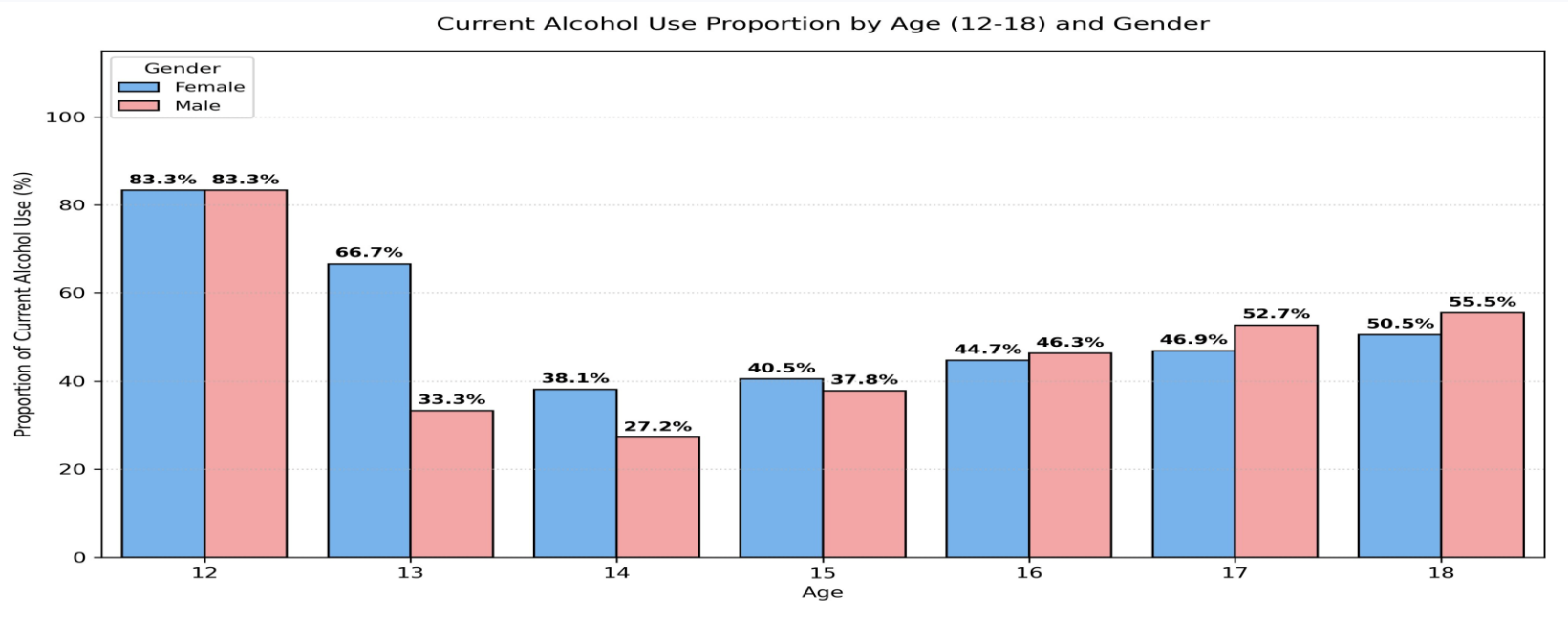
Age 17–18: Male overtakes Female (55.5% vs 50.5%)

Current Alcohol Use by Age & Gender

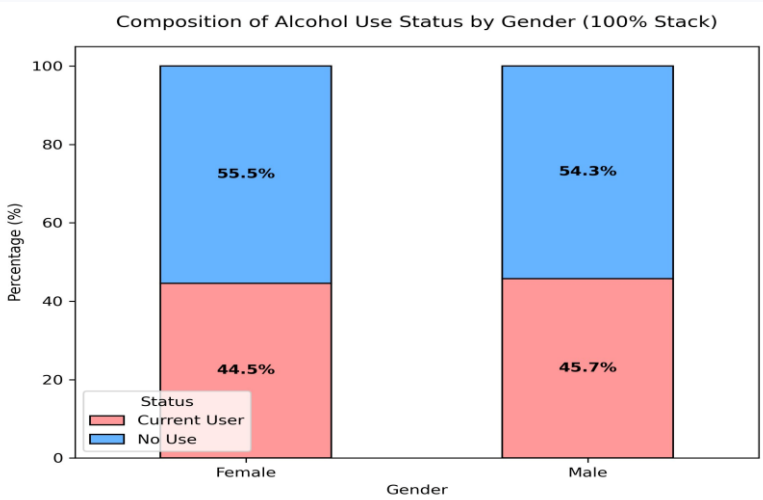
← Female leads here

⚠ Ages 12–13: n=12 & n=6 only — statistically unreliable

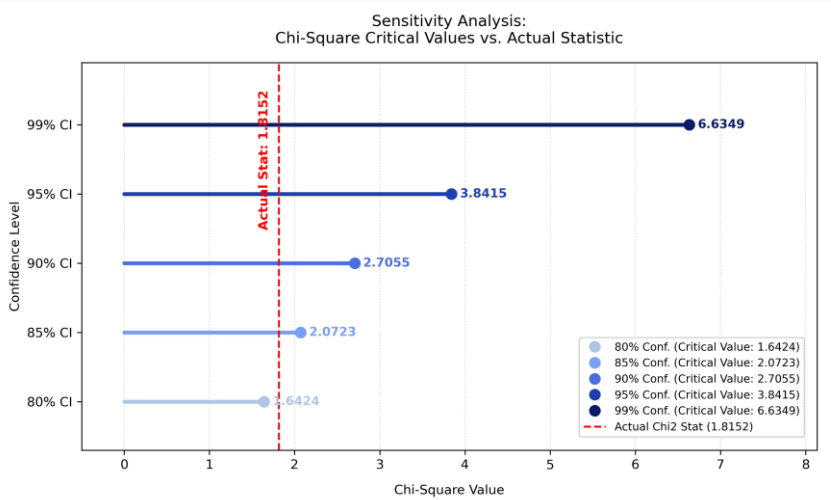
Male overtakes →



Overall Composition by Gender



Chi-Square Sensitivity Analysis



CONCLUSION No significant overall gender gap ($p = 0.178$). Age is the dominant driver. Gender differences emerge as a crossover pattern only within specific age cohorts.